

Total No. of Questions : 6]

SEAT No. :

P5505

[Total No. of Pages : 3

BE/Insem/Oct.-81
B.E. (Computer Engg.)
PRINCIPLES OF MODERN COMPILER DESIGN
(2012 Pattern) (Semester - I)

Time :1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*

- Q1)** a) Explain the role of Lexical analyzer in compilation process? [4]
b) What information is stored in symbol table? What are the issues in the design of symbol table? [4]
c) Explain the following [2]
i) Lexeme
ii) Lexical error

OR

- Q2)** a) Define the phase and a pass of a compiler? list machine dependent and machine independent phases of a compiler. [4]
b) Explain storage allocation strategies. [4]
c) Explain the meaning of following symbols in LEX. [2]
i) ? ii) []
iii) \$ iv) /

- Q3)** a) Computer first and follow sets for the following. [6]
Terminals = {id, nam, (,), ; , if, else,, \$}
Nonterminals = {S', S, L, C, E}
Start symbol = S

P.T.O.

- Rules =
- (0) $S' \rightarrow S \$$
 - (1) $S \rightarrow id \{L\};$
 - (2) $S \rightarrow if (E) S \text{ else } S$
 - (3) $L \rightarrow \epsilon$
 - (4) $L \rightarrow EC$
 - (5) $C \rightarrow \epsilon$
 - (6) $C \rightarrow ,EC$
 - (7) $E \rightarrow id$
 - (8) $E \rightarrow num$

- b) Explain type conversion. [2]
- c) Explain shift reduce and reduce - reduce conflict. [2]

OR

- Q4)** a) With respect to parsing explain the following terminologies. [4]
- i) Ambiguous grammar
 - ii) Follow rules

- b) Construct canonical LR parsing table for [6]

$S \rightarrow AB$

$A \rightarrow aA$

$A \rightarrow a$

$B \rightarrow Bb$

$B \rightarrow b$

- Q5)** a) Define synthesized and inherited attributes. [2]
- b) Explain intermediate code forms. [2]
 - c) Generate three address code for following. [6]

```

While (a < b)
{
    if (p < q and m > n)
    {
        x = x + 1;
    }
    else
        x = x - 1;
}

```

OR

Q6) a) Define the following terms with example [4]

i) Dependency graph

ii) L - attributed definition

b) Generate three address code, Quadruples, Triples and indirect triple for the following

$$S = (a + b) / (c - d) * (e + f) \quad [6]$$

